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GlassBox: Personalized Calibration for Privacy-Preserving with Explainable Exam Proctoring

GlassBox: Personalized Calibration for Privacy-Preserving and Explainable Risk Scoring in Online Exam Proctoring

Abstract

Online exams have become central to education, and cheating has grown with this shift. Existing facial recognition based tools apply one fixed detection standard to every student, causing documented unfairness toward students with darker skin tones and toward disabled students flagged for normal behavior. This makes fair, reliable exam monitoring a pressing problem.

We propose a browser based exam proctoring system using MediaPipe Face Landmarker for real time gaze tracking, combined with tab switching, focus loss, and copy paste signals. Since MediaPipe is a closed, pretrained model, we additionally fine tune an open source detector, RetinaFace, on diverse skin toned data and deploy it in the browser alongside MediaPipe, reducing detection bias at its source. Each student is further calibrated against their own baseline from the first 20 to 30 seconds of the exam, so remaining variation becomes part of their personal norm rather than a flagged deviation. These signals feed a logistic regression classifier, trained on the MSU OEP dataset, producing a transparent risk score for teachers, with disabled students receiving shifted thresholds rather than a separate rule. Compared to Proctorio and Examity, our system combines diversity aware detection, personalized calibration, and measured fairness through the FairFace dataset. A limitation since the system relies on browser-based signals rather than continuous video surveillance, a student using an undetected second device would not trigger any signal, regardless of teacher attentiveness.

Keywords: Exam proctoring system, Gaze tracking, Personalized calibration, Facial recognition, Algorithmic fairness, Explainable risk scoring.

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